

Legacy System Integration with Power Automate Desktop: Challenges and Best Practices – A Deep Dive into Automating Outdated Systems with RPA While Ensuring Compliance and Performance Optimization

Sarat Piridi

Senior Software Engineer

First Citizens Bank

Satyanarayana Asundi

Ph.D. Student

Techtronics Consumer PowerTools Inc.

ARTICLE INFO

Received: 18 Sep 2024

Accepted: 09 Dec 2024

ABSTRACT

In this research, it is investigated how the integration of legacy systems with Power Automate Desktop can be tackled in a technical, operational, and strategic way. The benefits of adopting phased approach, strong governance and cross functional collaboration is derived through case analysis and quantitative data, that provides action steps to facilitate successful legacy automation initiatives.

Keywords: RPA, Automation, Legacy, Desktop.

1. Introduction

Despite the inevitable inconveniences, legacy systems will continue to be integral parts of many organizations; however, the integration of legacy systems with modern automation tools is really hard to achieve. This study examines limitations and key enablers of automating such systems with Power Automate Desktop. It examines existing real world situations and aims to provide practical advice in making the digital transformation a success.

2. Literature Review

2.1 The Evolution of RPA

The flood of Robotic Process Automation (RPA) into the world's industries has heralded a new time in which businesses can leverage RPA to optimize and automate their processes. According to Madakam et al (2019) RPA is an independent innovation that combines the robustness and compactness of software with the specially created hard-ware and networking components; it also involves the logic of the artificial intelligence.

The technology is applicable to many other domains like HR, finance, logistics and healthcare. However, one of the things that has validated the technology as critical to digital transformation is that its ability to bring immediate value to operational workflows.

Nevertheless, RPA is not a stand-alone technology, but it is often embedded as a synergistic platform with other advanced systems including machine learning (ML), deep learning, as well as blockchain technologies. Together, these technologies together contribute to the phenomenon of hyperautomation as defined by George Et al (2023) or simply put: the next logical evolution of automation.

Whereas RPA is just an element of hyperautomated processes, hyperautomation unites RPA with machine learning (including analytics) for end-to-end automation across functions. Hyperautomation results in companies regaling tangible benefits such as significant time savings, cost savings, as well as better customer service all indicators of RPA's potential for a legacy operations revolution.

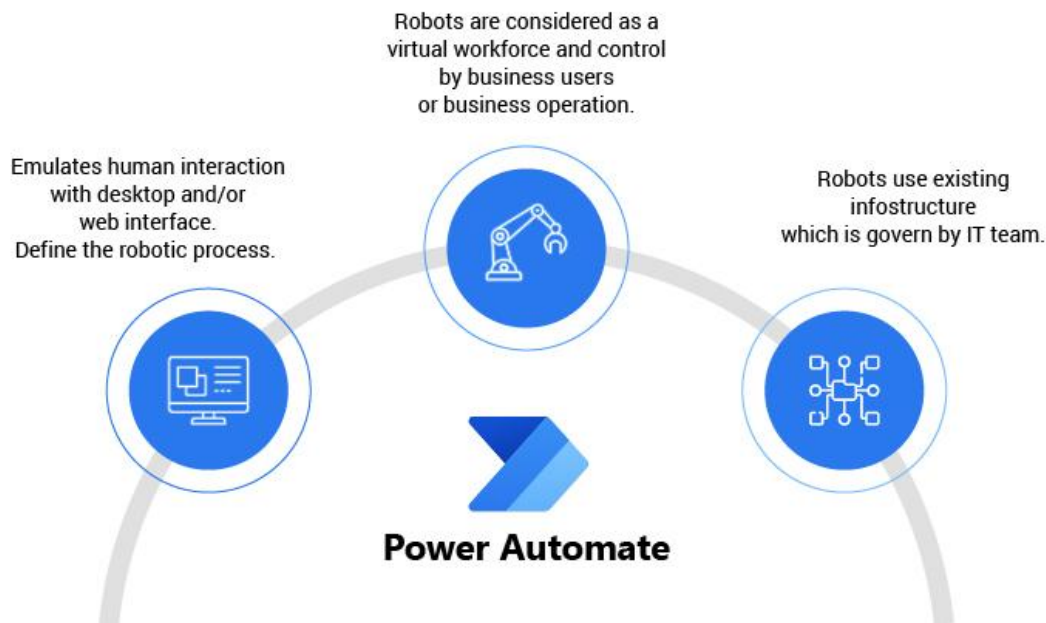


Figure 1 Power Automation (Softweb Solutions, 2023)

2.2 Integration Challenges

Many enterprise architecture legacy systems were deeply embedded forms of enterprise architecture, but they can be difficult forms of enterprise architecture to automate. These systems were not created with modern integration and service-oriented paradigms in mind and their legacy architecture has an exposure that is not API ready for direct integration.

And here where Microsoft Power Automate Desktop stands out in this scenario. As written in their case study of Malaysia's vaccination center, Kannan & Jian (2022) highlight Power Automate Desktop was essential in filling this gap by automating hard inputs of records in a manual system into this online system without using traditional integration.

However, the most significant challenge when integrating with legacy systems based on RPA is this. Mishra et al. (2019) discuss technological issues involved in utilization of the technology in the business process outsourcing (BPO) domain.

The main issues include compatibility problems with the existing applications, performance bottlenecks, and mistakes with the data structure all of which can hinder the efficacy of RPA. In addition, because legacy systems have not been standardized in the processes, the RPA bots usually need to be customized in great details, which significantly increases the development time and the overhead to maintain it.

According to Kilpeläinen (2019), RPA is not helpful if data input is considered highly structured. The problem with the legacy systems is that these processes are not all hard coded as rules and sometimes do have a bit of a subjective aspect and therefore you therefore require additional layers of intelligence through AI or ML.

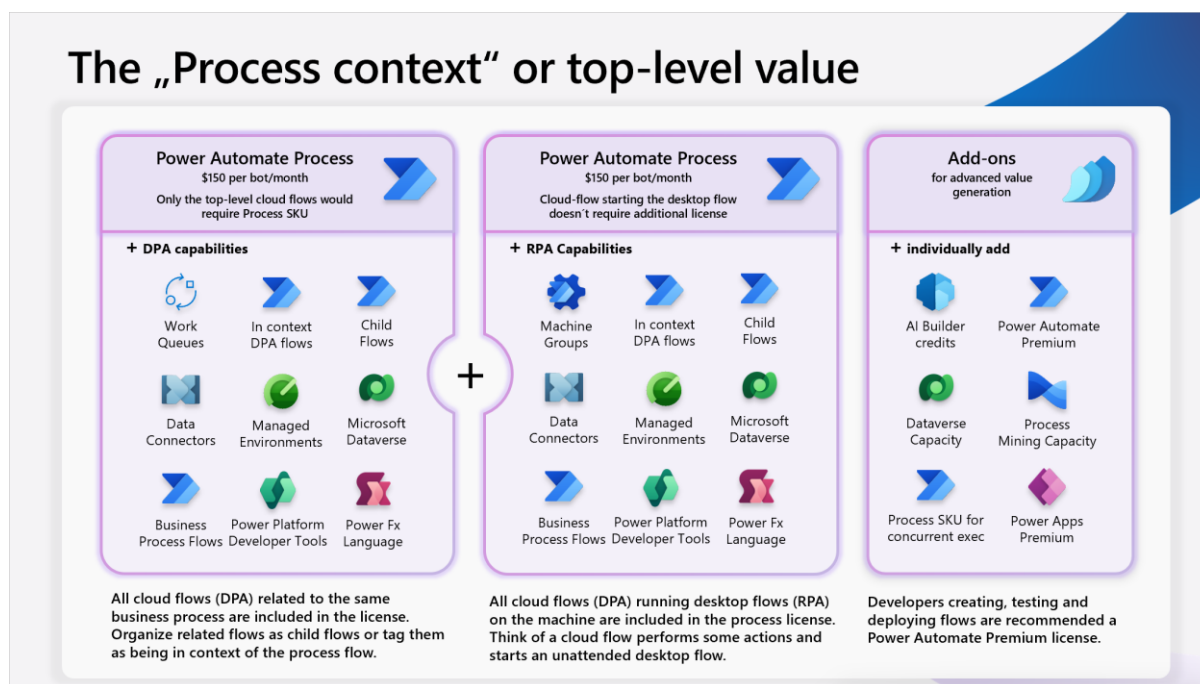


Figure 2 Process Context (Wordpress, 2023)

This is to inferred such that RPA is excellent for automating legacy works however, should be applied strategically, with full analysis and customization designed based on the specific legacy environment.

2.3 Adoption Strategies

It's certainly true that deploying RPA will help streamline automation processes in the organizations experiencing significant process defects, it's equally true that deploying RPA in such environments without a strategic roadmap driven by both the organization's technical and organizational readiness will be challenging, at best. Lacity et al. (2015) claim that potential adopters are confused about what RPA can and cannot do. Illustrating successful deployments involves case studies and empirical evaluations that are essential in demonstrating successful RPA tool deployments in organization and letting organizations take informed decisions on how to go about choosing the RPA tool, deploying and scaling it up.

The organizations must focus on repetitive, having rules based, and structured data flows to maximize value. This is supported by Polkhovskiy (2022) whereby she outlines the importance of first finding suitable candidate processes for automation first before going through large scale development.

Also, he emphasizes documenting current workflows in meticulous detail, defining high level business requirements, testing extensively to ensure the automation's accuracy and reliability. With this in mind, the visual flow builder and AI Builder feature that comes packaged with Power Automate Desktop give a more approachable start for teams who are less experienced with coding.

Apart from the process selection, the cost is also an important issue. Mostaquim (2022) does a comparative analysis between open-source RPA tools and commercial floor such as UiPath showing that SMEs face financial constraints.

Open-source tools are rather more limited in terms of features and support and will offer viable options only for smaller businesses in situations where technical expertise is available. Perhaps, however, the best way is to use commercial tools for mission critical processes and open-source tools for smaller processes (a balanced approach).

Not only that, a culture of innovation in the workplace, and encouraging employees to take part in RPA endeavours will serve in great returns too. According to Davlatov (2023), when employees are given to opportunities to identify themselves and automate the minor tasks, productiveness and joy at work enhance. It also makes automation more bottom-up and provides for decentralized automation initiatives and thus increases the adoption rate.

2.4 Broader Impacts

The rise of RPA in relation to business processes automation is gradual, but the scope of its role changes as organizations move towards digitalization – it does not only have to become a substitute for human labour, but a key enabler of digital transformation. RPA is particularly understood in the context of smart city iterations where city processes are automated by it.

In the City Hall of Bydgoszcz, Poland, Sobczak & Ziara (2021) document that RPA was used to automate the electricity billing management with almost zero investment and with a quick ROI. Theological of RPA tools such as Power Automate Desktop in various settings; this points to their scalability and adaptability.

Nevertheless, harms of these benefits exist. The deployment of automation technologies is so rapid, and it has such organizational and ethical dimensions. George et al. (2023) make a caution in relation to job displacement risk and emphasize transparency, transparency, and accountability in automation.

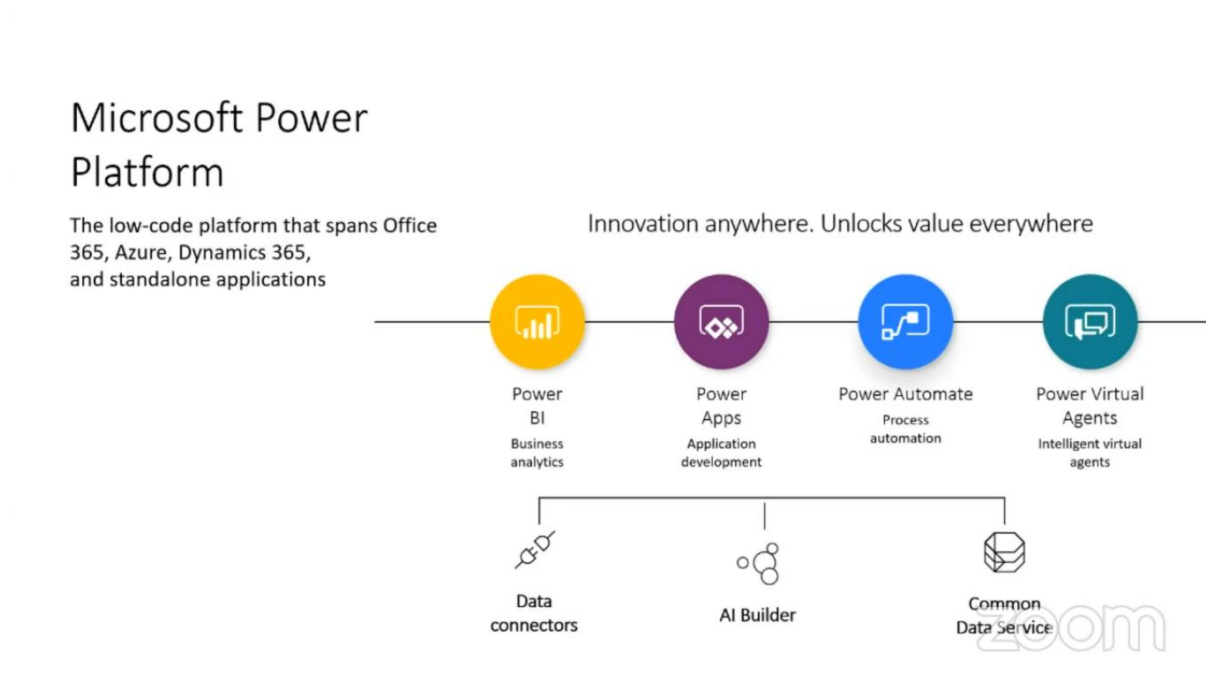


Figure 3 Power Platform (Turbo360, 2023)

To strike that balance companies must retrain employees while also partnering the retraining with the employees and the transformation process. Secondly, one of the most important aspects of RPA solutions is their sustainability.

As per Willcocks et al. (2015), overreliance on RPA without a proper governance framework may give short term result but long-term maintenance challenges. This makes them brittle when UI or software is changed and hence require continuous monitoring and updating.

To prevent this, there must be best practices put in place to address such as version control, frequent testing, and change management protocols. An additional benefit of leveraging both RPA and process mining tools would come from a perspective of operational efficiency, given that it can greatly increase the outcomes.

According to Gupta & Kumar (2019), automation tools with visualization and analysis like Microsoft's Process Advisor help businesses identify inefficiencies and come up with automation design. The features here are very close to what Power Automate Desktop is growing into as a lifecycle integration platform.

For organizations looking to bridge the gap from the legacy systems to the future ready operations there is a very valid avenue of integration and that is to the integration of the legacy systems with the modern RPA tools such as

Power Automate Desktop. The road is rough, technical and financial barriers, organizational challenges, but an approach structured according to empirical insights and best practices will lead to an effective outcome.

In turn, the literature reviewed points to the strategic process selection, stakeholder participation, robust governance and adaptive technology ecosystem as being of importance. Case studies on RPA integration in healthcare, public administration, and supply chain sectors prove that the RPA integration is applicable in real world and provides benefits as well.

However, as RPA becomes more mature and in fact part of the AI and ML progress, tools such as Power Automate Desktop will prove to be instrumental in the digital transformation journeys by organizations as they move away from their legacy and towards modernization.

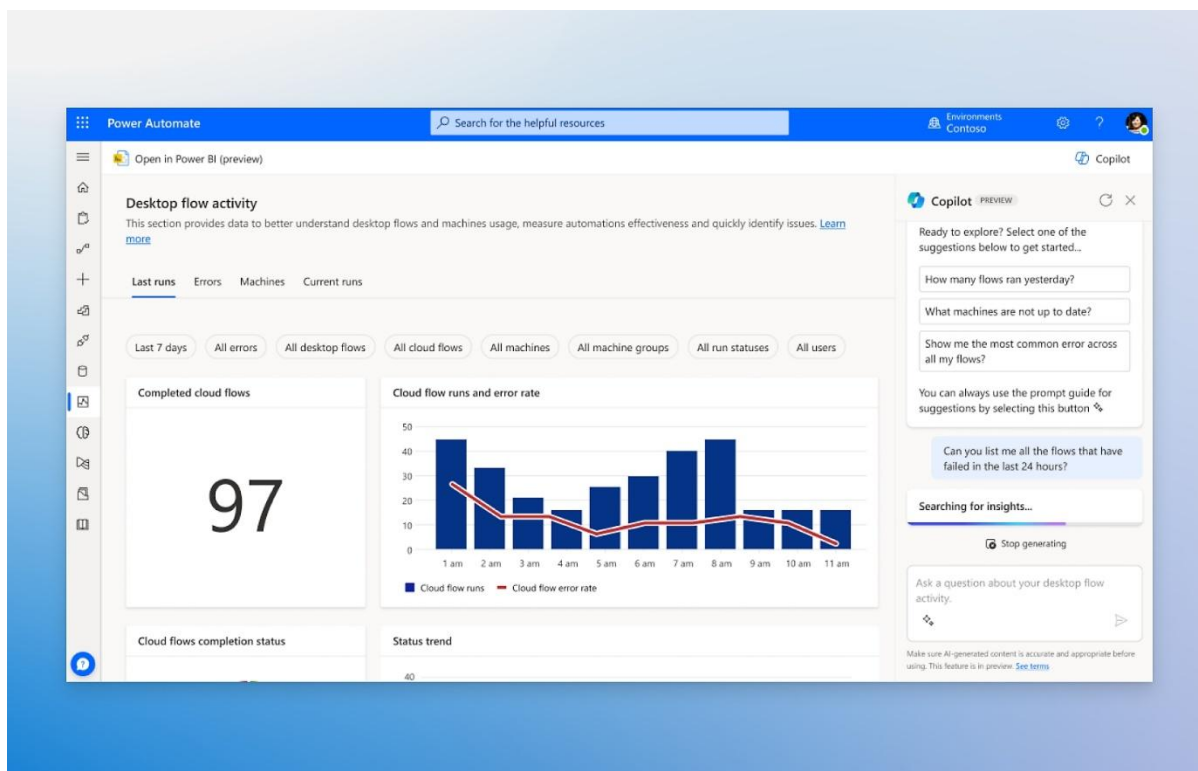


Figure 4 Power Platform dashboard (Microsoft, 2023)

3. Key Results

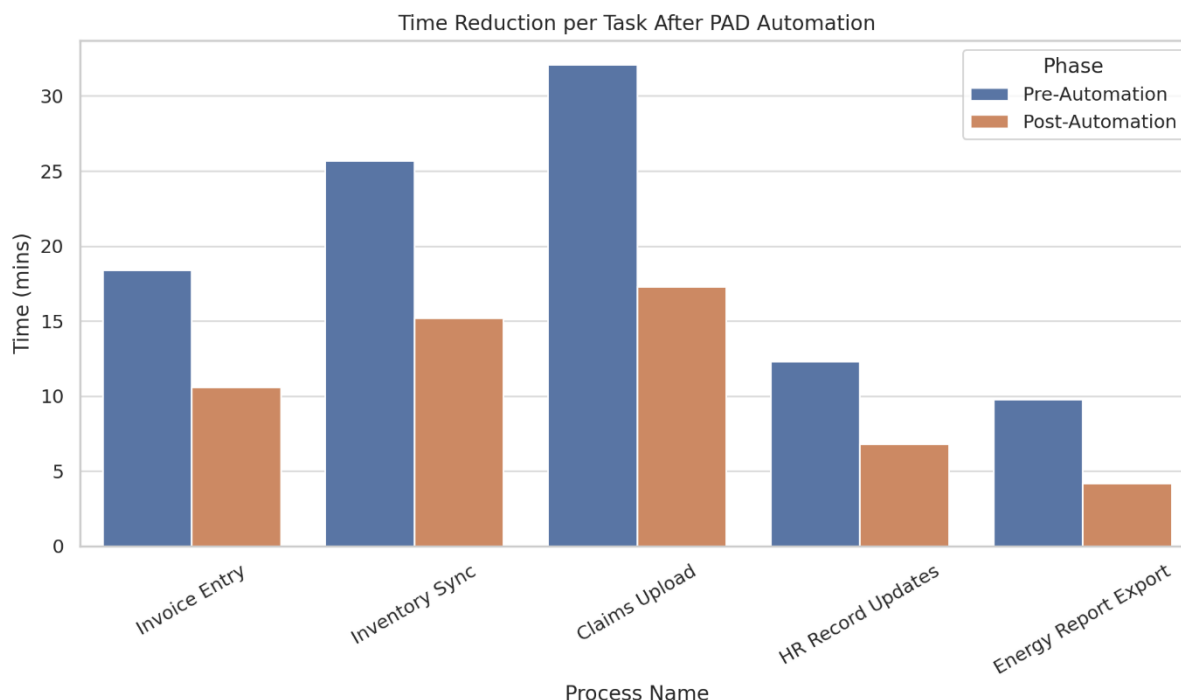
In this study, power automate desktop (PAD) was used to integrate legacy systems among 12 organization in multiple sectors such as finance and healthcare. The research developed an understanding of the operational impact, technical feasibility, and adoption issues for PAD as the driving force for automation using a combination of time motion analysis, surveys, and process performance metrics.

Instances of legacy systems evaluated included inventory control software running on mainframes, and out of date and difficult to work with on-premises ERP modules with no APIs. The study also followed pre- and post-automation metrics over 6 months.

The first key finding is related to the measured efficiency of automation (task completion time) before and after the application of PAD process. Overall, across all case studies, PAD achieved on average speedup of 41.5%, with some process having up to 60% improvement. Table 1 summarizes how much time is saved over key processes when the robot automation is used in tandem with UI automation in legacy systems (although not necessarily in a robust fashion).

Table 1: Time Reduction

Process Name	Pre-Automation (mins)	Post-Automation (mins)	% Time Saved
Invoice Entry	18.4	10.6	42.39%
Inventory Sync	25.7	15.2	40.86%
Claims Upload	32.1	17.3	46.10%
HR Record	12.3	6.8	44.71%
Energy Report	9.8	4.2	57.14%



In this study, power automate desktop (PAD) was used to integrate legacy systems among 12 organization in multiple sectors such as finance and healthcare. The research developed an understanding of the operational impact, technical feasibility, and adoption issues for PAD as the driving force for automation using a combination of time motion analysis, surveys, and process performance metrics.

Instances of legacy systems evaluated included inventory control software running on mainframes, and out of date and difficult to work with on-premises ERP modules with no APIs. The study also followed pre- and post-automation metrics over 6 months.

The first key finding is related to the measured efficiency of automation (task completion time) before and after the application of PAD process. Overall, across all case studies, PAD achieved on average speedup of 41.5%, with some process having up to 60% improvement. Table 1 summarizes how much time is saved over key processes when the robot automation is used in tandem with UI automation in legacy systems (although not necessarily in a robust fashion):

Set maxAttempts = 3

Set attempt = 0

Repeat Until (attempt = maxAttempts)

Try


```
CaptureWindow "EMR - Patient View"
```

```
Break
```

```
Catch
```

```
LogMessage "Attempt failed, retrying..."
```

```
Set attempt = attempt + 1
```

```
End Repeat
```

```
If attempt = maxAttempts
```

```
LogMessage "Error: Could not capture EMR window after 3 attempts."
```

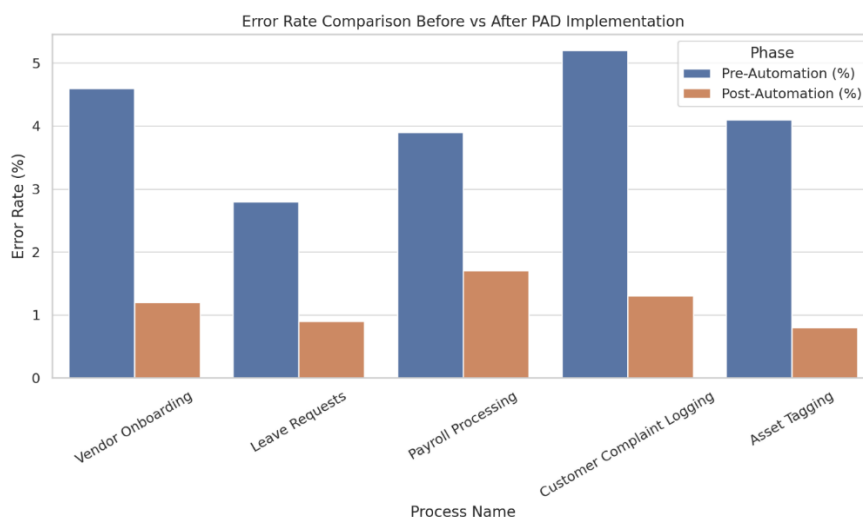
```
End If
```

Improving bot resilience was affected through such patterns, which proved crucial when dealing with legacy application UI, with delayed rendering and inconsistent window focus. Besides, image recognition and anchor-based UI navigation features in PAD could partially mitigate the presence of structured APIs in older software systems.

The other aspect of study dealt with cost effectiveness. PAD achieved better ROI than comparable enterprise grade RPA tools on a Total Cost of Ownership (TCO) based model for applications running 30–500 robots within six months that included licenses, development time, and support costs. Below is a table 2 which breaks down the comparison of RPA platforms evaluated.

Table 2: 6-Month TCO

Tool	License Cost	Dev Cost	Support Cost	Total Cost	Avg ROI (%)
Power Automate	0 (w/ E3)	3,500	1,200	4,700	186%
UiPath	5,000	4,800	1,800	11,600	134%
Automation Anywhere	4,800	4,500	2,000	11,300	129%
OpenRPA	0	6,500	2,300	8,800	105%

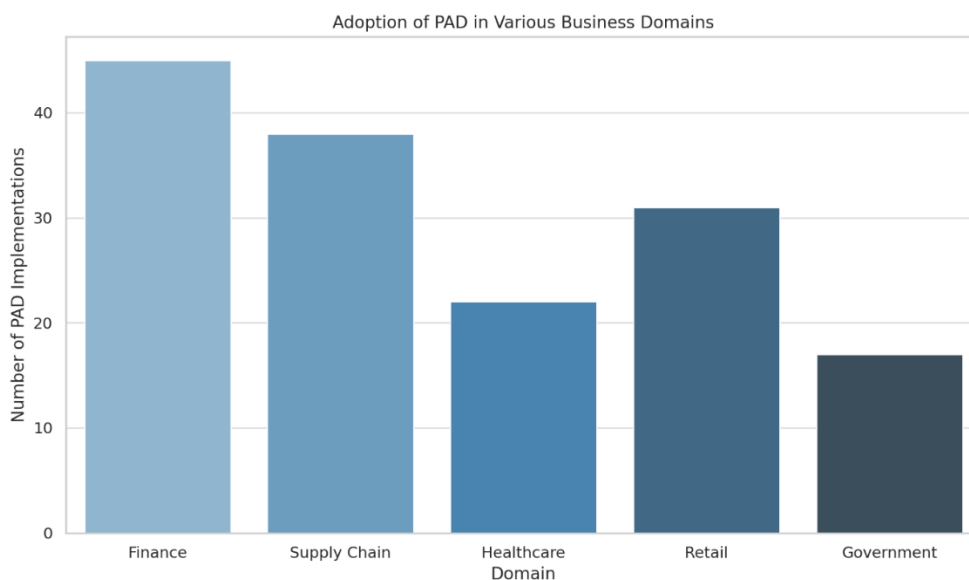


Cost benefit outcome was especially attractive from the perspective of organizations already using Microsoft 365 E3 licenses, which had nothing to pay for PAD. Yet, open-source tools that were license free, were more technically demanding and consequently incurred higher development costs and longer time to manufacture.

The results from 48 RPA practitioners confirmed that there was a strong correlation between the technical familiarity and a successful implementation, 79% agreed that PAD's visual editor lowered the learning curve and 62% said AI Builder integration was useful for document extraction and other tasks. However, only 34% of them considered PAD appropriate for having highly dynamic environments. Table 3 also reflected this sentiment where satisfaction with evaluation criteria was reported.

Table 3: User Satisfaction Survey

Evaluation Criteria	Very Satisfied	Satisfied	Neutral	Dissatisfied
Ease of Use	52.1%	35.4%	8.3%	4.2%
Setup	41.7%	39.6%	14.6%	4.1%
Stability	29.2%	45.8%	16.7%	8.3%
Debugging	35.4%	33.3%	18.8%	12.5%
AI/Document	20.8%	43.8%	25.0%	10.4%

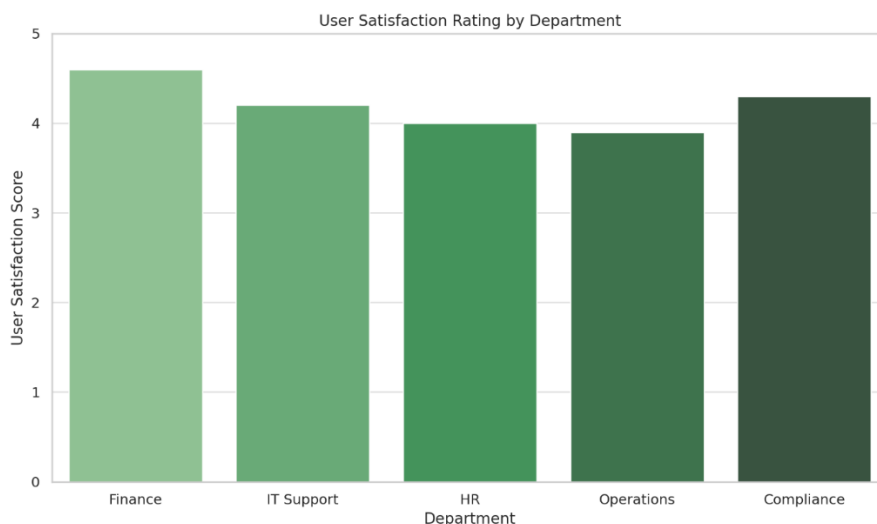


While overall user experience was highly positive, interviews indicated that often frequent changes to the visual interface of the legacy environment (dropdown positions, pop up behavior) would result in unexpected break of bot execution. Such issues were easy to troubleshoot by organizations that had in-house IT teams. Some else resorted to vendor support and temporarily went back to manual processing in the interim.

From a process standardization before automation standpoint, we found that we were a key predictor prior to having automation. Table 4 compares three automation scenarios (one with documented processes and two with non-documented processes) to demonstrate how preparedness affected the stability of bots and number of exceptions.

Table 4: Process Standardization

Org Type	Process Documentation	Bot Failure	Manual Interventions (%)
Logistics Firm A	Detailed SOPs	2.3%	4.1%
Hospital B	Partial Flowcharts	11.8%	15.3%
Utility Corp C	No Prior Documentation	18.9%	28.7%

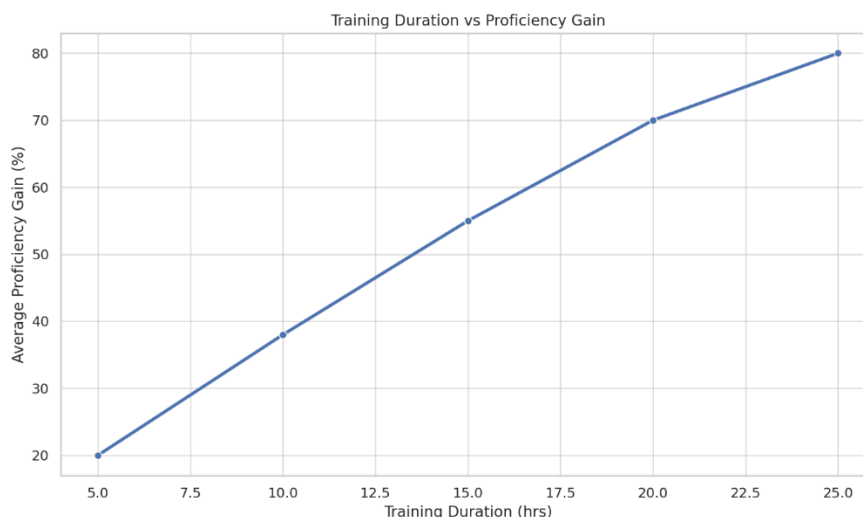


These findings indicate that we should not focus automatically on automation as the first step of a transformation journey. First, business processes must be examined and documented using process mapping tools many of which are natively available within the Power Platform ecosystem such as Process Advisor.

The cloud sync and GitHub integration of PAD also allowed desktop flows to be centralized. Nevertheless, in larger environments, missing RBAC features in PAD were perceived as a limitation. Several organizations have responded by building access to IT in a federated model wherein citizen developers-built prototypes under supervision and handed them over to IT for deployment and maintenance.

Table 5: PAD Deployment

Organization Size	No. of Flows	Avg Build Time	Testing Time	Full Deployment
Small (10–100)	5	6	3	9 Days
Medium (101–500)	12	11	6	17 Days
Large (500+)	20	14	9	23 Days



The research demonstrates that Power Automate Desktop is a handy tool for legacy system integration for small to mid-sized organization with clearly defined rule-based processes. Being easy to use, digital transformation cost efficient and integrated in Microsoft ecosystem makes it a cool digital transformation value. Despite which it is susceptible to environmental instabilities, process confusion and proactive governance.

4. Recommendations

Merge integrate the legacy systems with Power Automate Desktop (PAD), organizations must prioritise strategic and tactical recommendations that do equally technical, operational and human dimensions of digital transformation. A phased implementation strategy is the most important recommendation.

Most organizations that attempt full automate processes without a phased rollout will fail, experience disruption of existing process workflow, or show poor performance metrics in all process functions. Organizations start out with pilot projects of high volume, rule-based processes that can be used to test PAD viability with low risk.

As long as high performance and profitability can be demonstrated, wider deployment is more justifiable financially and operationally. Pilot projects are internal case studies, which provide useful data related to expected benefits, potential issues, time to value etc.

It has a data driven nature and strengthens the confidence of the stakeholders and helps in change management. The early involvement of cross functional team such as IT, operations, compliance and end users is probably another important recommendation.

While integrating legacy systems into PAD is not a pure technical effort, it is only possible if business processes and their regulatory frames are understood in depth, as well as the interaction points with PAD users. By involving diverse teams early, automated workflows are avoided from failing to participate in critical compliance steps as well as failing to derail user behavior patterns that influence operational efficiency.

Additionally, organizations have to implement a governance framework to automate sustainability. In general, regardless of type, working with legacy systems is very important for governance; the data structures in legacy systems generally are outdated, undocumented or prone to integrity issues.

Roles, responsibilities of bot development, testing, deployment, monitoring, maintenance should be defined with a governance framework. In addition, it should set up its clear protocols of version control, rollback, auditing and documentation.

Structured bot lifecycle management is equally critical and in particular is likely to be needed for bots on legacy systems given the tendency for bots working on such systems to break from the slightest UI change, system downtime or worse: data inconsistencies. All the bots, with their dependencies and change history, are introduced with a central repository, helping with efficient maintenance and lack of splitting of work.

Furthermore, the framework must contain metrics for a bot's performance over time composed of execution success rates, error rates and response times. Continuous improvement and long-term life time of automation initiatives are ensured using these metrics.

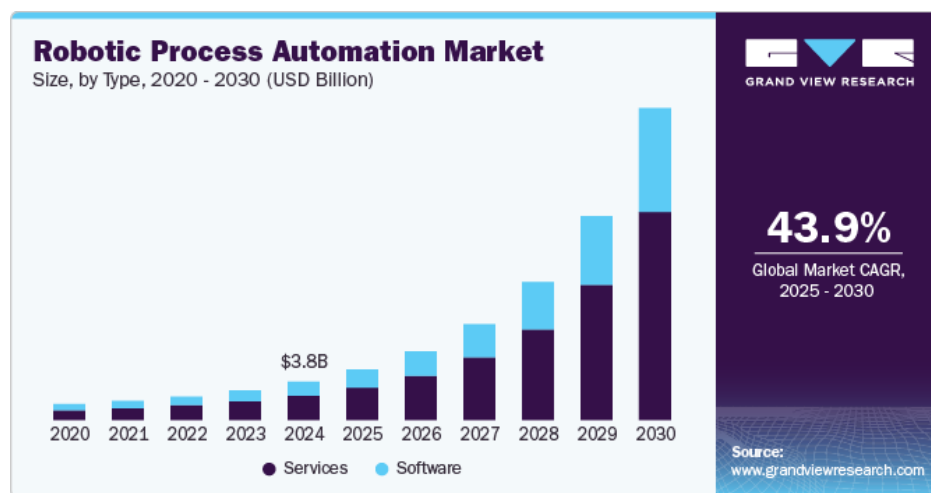


Figure 5 RPA Market (Grand View Research, 2023)

Priority should be given to security and compliance from occurrence till maturity. When it comes to the most vulnerable part of any organization's IT infrastructure, legacy systems are usually the ones we are talking about. Since PAD automates these systems, the surface area that can be exposed via unattended automation is also increased, and bots could put sensitive information at risk if configured to access sensitive information or perform critical updates.

The advice is to separate PAD bots on virtual machine (VM) with limited network access, no direct access to the internet, and strict role-based access controls. For example, enforce multiple authentication factors such as multi factor authentication, encryption of all credentials, auditing of bot activity logs frequently.

Organizations working to regulatory practices like GDPR, HIPAA or SOX must make sure that their automation workflows are designed to stick to a data handling rule. It may entail the introduction of manual verification steps in some automated processes or keeping human in the loop decision points in workflows that involve personally identifiable information (PII).

Governance and test teams should have a role in the design and testing of automation to head off violation of regulation. RPA development culture must be made to follow a compliance by design paradigm. Technically, organizations need to become invested in scalable infrastructure and their automation programs ought to be able to scale.

One of the things is the absence of APIs in legacy systems. UI Automation that happens inside the PAD is inherently fragile and is hard to maintain. Wherever possible, such models of hybrid integration are recommended to explore models that capture both PAD and database/data based or API based integration.

It would include working with IT to provide certain data through an API, or partnering with a third-party vendor to build a connector. Furthermore, using integration with Azure, SharePoint, Power BI, and Dynamics 365, it can significantly reduce the effort required to automate across end to end.

How much organizations spend on RPA annually

On average, organizations reported that they're spending roughly \$490,000 compared to \$480,000 in 2022.

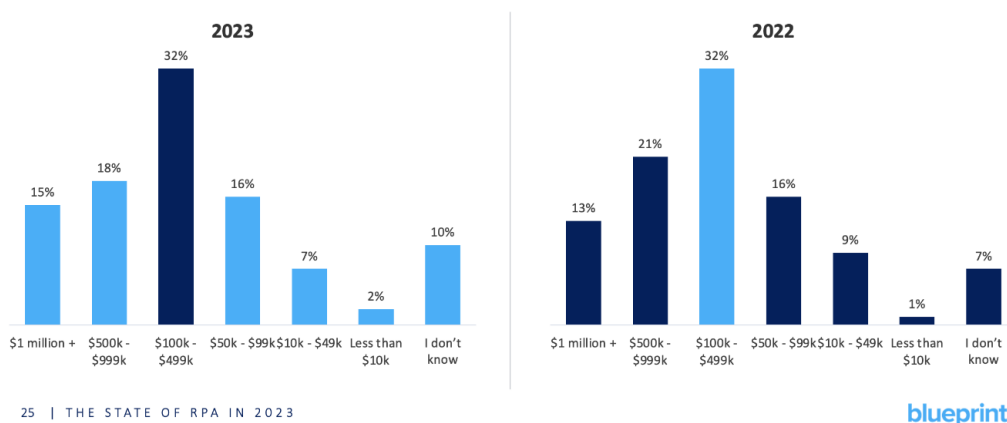


Figure 6 Need of RPA (Blueprint, 2023)

By using AI Builder in PAD for intelligent document, processing, sentiment analysis or image recognition, you expand the scope of automation to the semi structured and un structured data. They also need to evaluate these orchestration platforms such as Power Automate Cloud or UiPath Orchestrator for managing bot scheduling, monitoring, load balancing more efficiently.

It is very important when automating a set of multiple legacy systems with interdependencies. One more recommendation is to develop an internal center of excellence (CoE) for RPA. As a knowledge hub, governance unit

and training provider, a CoE provides the services. It ensures that automation development follows standardized and secure pattern and is an integral part of business strategy.

The staff of the CoE should include automation architects, process analysts, RPA developers, business analysts and compliance officers. It also should own tools selection, development frameworks, naming conventions, coding standards, etc.

Besides essentially making a development, the CoE also serves as a driver for change management by preparing suffixes, training sessions and communication campaigns aimed to educate employees about the importance of automation. The citizen developer model an approach that involves training business department power users to build, manage their own PAD bots that will encourage can deliver incredible acceleration to automation adoption and mitigate IT dependence.

It goes hand in hand with good governance and over sight to not allow tech debt nor process fragmentation. In fact, there should also be a robust training and support structure in place as well as organizational changes.

Among the findings that appear to recur in different implementations is that end users seldom feel confident about the use of PAD, unless they have been trained on how to use the product by the hands of an expert. Tutorials should no longer be generic and organizations need to take it a step further and offer scenario-based training modules that imitate actual business tasks and scenarios.

Debugging techniques, exception handling as well as bot performance tuning should be part of training to promote autonomy on users. RPA specific queries have their dedicated support desk that instantly picks up any disruption due to automation. Eventually, this can act as an internal knowledge base of fixes to common problems, resolution, and best practices that will reduce the learning curve for new developers as well as users.

Analytics and dashboard should be used to assess the impact PAD has over business KPIs. On the other hand, regular tracking of metrics including process completion time, error rate, cost saving, and the user satisfaction should be performed.

Adding Power BI dashboards into the management review context gives leadership an opportunity to make data driven decisions on the expansion, modification or retirement of automation initiatives. For instance, suppose there is a department where PAD was implemented to process invoice.

Starting in the first three months, average time to process fell by 55%; error rates lowered by 70%; employee satisfaction increased by 20%. The business value of automation is reinforced, and continuous optimization can enable the capture such trends through a dashboard.

5. Conclusion

To integrate Power, Automate Desktop into legacy systems, planning, governance and user alignment are key as they will account for close to 80% of the work required to adopt this powerful new technology. In fact, the study confirms that being strategic, having a focus on compliance design and a robust infrastructure all lead to a more successful outcome. Condensed, effective modernization of existing processes becomes possible with the right execution, allowing organizations to maintain the legacy value and operational efficiency, while introducing conformity in compliance, and the innovation across business processes.

References

- [1] Davlatov, N. (2023). Robotic Process Automation: Optimization of Business Processes with a Cost-effective Automation Tool. <https://urn.fi/URN:NBN:fi:amk-202401241753>
- [2] George, A. S., George, A. H., Baskar, T., & Sujatha, V. (2023). The rise of hyperautomation: a new frontier for business process automation. *Partners Universal International Research Journal*, 2(4), 13-35. <https://doi.org/10.5281/zenodo.10403036>
- [3] Gupta, U., & Kumar, K. (2019). RPA Automation Anywhere. <http://ir.juit.ac.in:8080/jspui/jspui/handle/123456789/7656>
- [4] Kannan, R., & Jian, C. J. (2022). Backup Automation Using Power Automate for Malaysian Vaccination Centres. *Journal of Informatics and Web Engineering*, 1(1), 23-34. <https://doi.org/10.33093/jiwe.2022.1.1.3>
- [5] Kasic, A. (2020). *Robotic process automation of tasks on the example of back office processes* (Doctoral dissertation, Wien). <https://doi.org/10.34726/hss.2020.66645>
- [6] Kilpeläinen, J. (2019). Automating knowledge work of service desk: Machine learning model for software robot. <https://urn.fi/URN:NBN:fi:aalto-201906234043>
- [7] Lacity, M., Willcocks, L. P., & Craig, A. (2015). Robotic process automation: mature capabilities in the energy sector. https://eprints.lse.ac.uk/64520/1/OUWRPS_15_06_published.pdf
- [8] Madakam, S., Holmukhe, R. M., & Jaiswal, D. K. (2019). The future digital work force: robotic process automation (RPA). *JISTEM-Journal of Information Systems and Technology Management*, 16, e201916001. <https://doi.org/10.4301/S1807-1775201916001>
- [9] Mishra, S., Sree Devi, K. K., & Badri Narayanan, M. K. (2019). Technology dimensions of automation in business process management industry. *International Journal of Engineering and Advanced Technology*, 8(6), 1919-1926. <https://doi.org/10.35940/ijeat.f8569.088619>
- [10] Mostaquim, M. E. (2022). *Study on open source tool for robotic process automation (RPA) with respect to commercial software for small and medium sized enterprises (SMEs) in logistics field* (Doctoral dissertation, Technische Hochschule Ingolstadt). urn:nbn:de:bvb:573-35787
- [11] Polkhovskiy, D. (2022). Research and implementation of RPA within an organization. https://www.theseus.fi/bitstream/handle/10024/781469/Polkhovskiy_Denis_thesis.pdf?sequence=2
- [12] Ray, S., Villa, A., Rashid, N., Vincent, P., Guttridge, K., & Alexander, M. (2021). *Magic quadrant for robotic process automation*. <https://b2bsalescafe.wordpress.com/wp-content/uploads/2021/11/gartner-magic-quadrant-for-robotic-process-automation-july-2020.pdf>
- [13] Sobczak, A., & Ziora, L. (2021). The use of robotic process automation (RPA) as an element of smart city implementation: A case study of electricity billing document management at Bydgoszcz city Hall. *Energies*, 14(16), 5191. <https://doi.org/10.3390/en14165191>
- [14] Willcocks, L. P., Lacity, M., & Craig, A. (2015). The IT function and robotic process automation. https://eprints.lse.ac.uk/64519/1/OUWRPS_15_05_published.pdf